

Project 1

M1. Meta-Analysis Automation Kit: Full Elaboration

1. Methodological work (the "right way" thinking)

- **Research question framing:** PICO (Population, Intervention, Comparator, Outcome) and inclusion/exclusion criteria
- **PRISMA 2020 compliance:** identification → screening → eligibility → included, with reasons for exclusion
- **Effect size choice:** SMD (Hedges' g), mean difference, odds ratio, risk ratio or log-HR, matched to the data type
- **Model choice rule:** fixed-effect only if heterogeneity is low and studies are similar. Otherwise random-effects (DerSimonian-Laird or REML)
- **Heterogeneity logic:** I^2 (<25% low, 25-75% moderate, >75% high), Q test p-value, τ^2
- **Sensitivity analysis:** leave-one-out, to show no single study drives the result
- **Subgroup / meta-regression:** only if there are enough studies (rule of thumb: ≥ 10)

2. Technical work (the "automation" layer)

- CSV validation: missing values, duplicate studies, wrong variance, negative sample sizes
- Python pipeline: pandas, numpy, scipy, statsmodels, matplotlib
- Hand-coded formulas (inverse-variance weights, pooled estimate, CI, Q, I^2), so the maths is transparent and checkable
- Auto model selector: if $I^2 >$ threshold, switch to random-effects, and log the reason
- Reproducibility: fixed seed, one function per step, exportable results (CSV, PNG, SVG)
- Optional R cross-check (metafor) to prove the numbers match

3. Analytical work (the "interpretation" layer)

- Pooled effect, 95% CI, p-value, prediction interval
- Plain-language reading: "Effect is significant, but heterogeneity is high, so interpret with caution"
- Publication bias check: funnel plot symmetry plus Egger's test ($p < 0.05$ means possible asymmetry)
- Optional trim-and-fill to estimate the impact of missing studies
- Certainty hint (GRADE-style note): risk of bias, inconsistency, imprecision

4. Report and visualization work (the "looks professional" layer)

- **PRISMA flow diagram:** clean boxes, journal-style, auto-filled with your counts
- **Forest plot:** study squares sized by weight, diamond for pooled effect, I^2 printed on the plot
- **Funnel plot:** with pseudo-confidence limits
- **Pooled result table:** one table for fixed vs random, ready to paste in a thesis
- **Auto results paragraph** in thesis style, with numbers filled in
- **Consistent palette and fonts**, 300 DPI export, so figures fit a Word or LaTeX thesis

How you act as a proxy / fill the skill gap

| Client's gap | What you deliver |

|---|---|

Doesn't know which model	Auto decision plus a one-line justification
Can't code or use Python, R	A finished figure pack and tables
Weak at writing results	A ready results section and a methods note
Fears supervisor rejection	PRISMA-compliant, reproducible output

Client deliverables package

1. Figures: PRISMA, forest, funnel (PNG and vector)
2. Pooled result table (Excel or Word-ready)
3. Results and methods paragraph (editable)
4. A short "how to interpret" note, so the student can defend it in viva

Next programming and design parts.

Project 2 **M2. Clinical Statistics Report Generator: full scope**

****The gap you fill:**** The doctor knows the clinical question. They do not know which test fits, how to check assumptions, or how to write results in journal style. You act as their statistician, analyst, and designer in one.

****1. Methodological work (choosing the right method)****

- Read the research question and decide the study design: cohort, case-control, or cross-sectional.
- Match the outcome type to the test:
 - Continuous, normal: t-test or ANOVA
 - Continuous, skewed: Mann-Whitney or Kruskal-Wallis
 - Categorical: Chi-square or Fisher's exact
 - Binary outcome: logistic regression
 - Time-to-event: Kaplan-Meier, log-rank, Cox model
- Check assumptions (Shapiro-Wilk for normality, Levene for equal variance, proportional hazards for Cox, multicollinearity by VIF) and record why each test was chosen.
- Decide how to handle missing data: delete, impute, or report as missing.
- Choose confounders for adjusted models, and keep events-per-variable in mind.

****2. Technical work (the pipeline)****

- Excel upload, column detection, fixing wrong data types, outliers, and impossible values (age 250).
- Automatic Table 1 with mean (SD) or median (IQR), n (%), and p-values.

- Test selector function: it reads each variable and applies the right test.
- Models: logistic regression, Kaplan-Meier, Cox, ROC with AUC and the best cutoff (Youden index).
- Output exported to Excel, Word, and PDF.

****3. Analytical work (interpreting, not only calculating)****

- Odds ratio and hazard ratio with 95% CI, explained in plain words ("smokers had 2.4 times higher odds").
- Crude vs adjusted results side by side.
- AUC, sensitivity, and specificity meaning for clinical use.
- A limitations section: small sample, missing data, and no causal claim.

****4. Report and visualization work (the polished look)****

- Journal-style Table 1 with clean fonts, aligned decimals, and footnotes.
- Forest plot of odds ratios.
- Kaplan-Meier curves with a risk table, confidence bands, and a log-rank p-value.
- ROC curve with the AUC label and the cutoff point marked.
- Consistent colour palette, 300 DPI figures, and captions in journal style.
- Auto-written Methods and Results draft, for example: "A total of 240 patients were included. Median age was 54 years (IQR 45–63)..."

****What the client receives****

- Cleaned dataset and a data-cleaning log
- Table 1 and results tables
- Four to five publication-ready figures
- Methods and Results draft (Word)
- A short "how to read this" note for the doctor

****How to pitch it on your portfolio:**** show the same dummy dataset going from raw Excel to a finished report, with before and after screenshots. This shows the doctor that waiting weeks and paying a statistician is no longer needed.

Next hol programming and design

Project 3:

****M3. Diagnostic Test Evaluation Suite: full scope****

****The gap I fill up:**** The doctor has a new test (a score, a marker, an AI tool) and a gold-standard result. They do not know how to prove the test is good in the way journals and reviewers expect. You deliver the proof and the paperwork.

****1. Methodological work****

- Follow the STARD checklist, so the report matches what journals ask for.
- Use exact (Clopper-Pearson) or Wilson confidence intervals, because the normal approximation fails for small samples and for values near 100%.
- Add positive and negative likelihood ratios.
- Recalculate PPV and NPV at the real-world prevalence. A study sample often has more disease than a normal clinic, so the original PPV and NPV can mislead.
- Compare two tests with DeLong (AUC, unpaired view of discrimination) and McNemar (paired accuracy on the same patients).

****2. Technical work****

- ``sklearn.metrics.roc_curve`` for the ROC and AUC.
- ``statsmodels.proportion_confint`` for the sensitivity and specificity CIs.
- A small DeLong function written by you, since it is not built into sklearn.
- Bootstrap (1,000+ resamples) for the confidence interval of the optimal cut-off.
- Output exported to PDF, Word, and Excel.

****3. Analytical work****

- Check the reference standard: is it truly gold, and was it applied to everyone?
- Look for spectrum bias: were only very sick and very healthy patients included?
- Choose the cut-off by Youden index or by cost. If missing a disease is worse than a false alarm, pick a cut-off with higher sensitivity.
- Add decision curve analysis if the reviewer asks "is it clinically useful?"
- Explain results in plain words: "At 8% prevalence, a positive result means only about 1 in 3 are truly sick."

****4. Visualization work****

- ROC curve with a confidence band and the optimal point marked.
- Two-test ROC overlay with the DeLong p-value.
- Confusion matrix heatmap with counts and percentages.
- Fagan nomogram, which shows how pre-test probability changes to post-test probability.
- One-page PDF summary with the key numbers, curves, and a verdict.

****5. Proxy extras (what makes clients pay)****

- STARD flow diagram: eligible patients, excluded, tested, final analysis.
- Ready Methods paragraph for the paper.
- Reviewer-reply wording, for example: "We thank the reviewer. We now report 95% exact CIs and prevalence-adjusted predictive values (Table 2)."

****What the client receives****

- Complete accuracy table with 95% CIs
- Four to five publication-ready figures
- STARD flow diagram and checklist
- Methods and Results draft
- One-page summary and reviewer-comment replies

****Portfolio tip:**** show one dummy dataset where the PPV changes sharply between 50% and 5% prevalence. It shows clearly why your work catches an error that many doctors miss.

M4. Research Proposal & Sample Size Planner: full scope

The gap I fill up: The researcher has a topic or idea, but no clear question, no sample size, and no plan a supervisor or ethics board will accept. You turn the idea into a defensible proposal.

1. Methodological work

Shape the idea into a PICO question (Population, Intervention, Comparator, Outcome), then check it with FINER (Feasible, Interesting, Novel, Ethical, Relevant).

Match the design to the question: RCT, cohort, case-control, cross-sectional, or diagnostic study.

Use the right formula for the outcome:

Two means

Two proportions

Correlation

Survival (number of events needed)

Diagnostic accuracy

Use 80% or 90% power and 5% alpha, then inflate for dropout (for example, divide by 0.85 for 15% loss).

2. Technical work

Biopython Entrez to pull PubMed titles and counts (needs internet in Colab).

statsmodels power functions for sample size.

pandas for keyword counts by year and for the budget table.

Plotly or matplotlib for the Gantt chart and the power curve.

3. Analytical work

Gap analysis: count keyword combinations by year to show what is studied and what is missing.

Effect size must come from a cited paper, not a guess. Write the source in the justification.

Sensitivity table: show how the sample size grows if the true effect is smaller (for example, 20% or 30% smaller).

Feasibility check: can the sample be recruited in the time and budget available?

4. Visualization work

PubMed publications-per-year bar chart (shows the gap).

Power curve (sample size vs power).

Gantt timeline of the study phases.

Clean budget table with totals.

5. Proxy extras (what saves the client time)

Sample size justification paragraph, ready to paste.

Statistical analysis plan: which test for which outcome, and how missing data is handled.

Ethics section draft: consent, confidentiality, risk and benefit.

Short feasibility summary for the supervisor.

What the client receives

Final PICO question and study design

Sample size with formula, assumptions, and sensitivity table

Four visuals, ready for the proposal

Proposal draft with the statistical plan and ethics text

Why this works as a skill-gap service (all four projects, M1 to M4)

The client brings raw data or an idea. You deliver:

The correct method

Numbers that can be defended

Journal-quality figures

Written text

So supervisors and reviewers have very little to send back.

Portfolio tip: show one sensitivity table where a smaller effect size doubles the required sample.

It proves that you protect the client from an underpowered study.